Learning to Upsample and Upmix Audio in the Latent Domain

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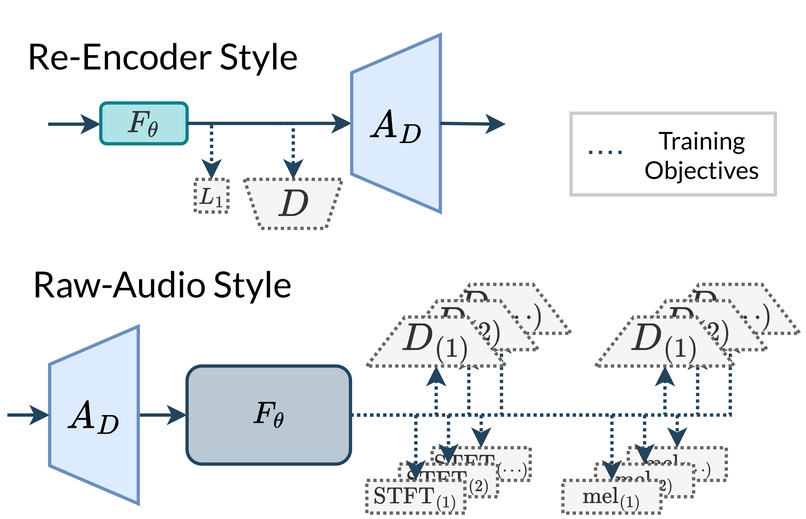
Motivation

- **Neural audio autoencoders** power compression & generation.
  - Compact latents preserving perceptual information.
- Yet, most audio is processed as a waveform or a spectrogram.
  - **Poor integration** with autoencoder pipelines.
  - Compute & memory **inefficiency**.
  - Sensitive, **tuning-heavy** audio objectives.
- Existing latent-based approaches are costly (e.g. LLMs, DiTs).

Can we process audio efficiently, directly in the latent space?

Re-Encoder Approach

- **Process audio entirely in the latent space** of a pre-trained autoencoder (A_E, A_D), during **both** training and inference.



- Seamless integration into pipelines.
- No decoding for intermediate processing.
- Simple training objectives.
- Low compute requirements.
- Autoencoder agnostic.

Latent Space Training

- Audio $\boldsymbol{x} \in \mathbb{R}^L$ encoded to compact latent $\boldsymbol{z} = A_E(\boldsymbol{x}) \in \mathbb{R}^{C \times T}$.
- Train task-specific predictive model F_θ estimating target $\boldsymbol{z}_{tgt}$ given input $\boldsymbol{z}_{in}$ and optional condition $\boldsymbol{c}$.

$$L_{\text{rec}} = \mathbb{E}_{\boldsymbol{x}_{\text{in}}, \boldsymbol{x}_{\text{tgt}}, \boldsymbol{c}} \left[\left\| A_E(\boldsymbol{x}_{\text{tgt}}) - F_\theta(A_E(\boldsymbol{x}_{\text{in}}), \boldsymbol{c}) \right\|_1 \right].$$

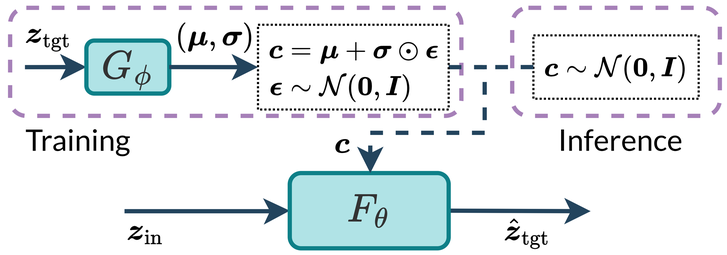
- Latent proximity maps to waveform perceptual similarity.
- Autoencoder kept frozen.

Latent Space Discriminator

- We employ a **latent adversarial discriminator** to better match the target distribution.
- **Adversarial** & **feature-matching** training objectives.

Latent Space Variational Encoder

- We want to model a distribution over potential target outputs.
- We condition on a low-dimensional vector $\boldsymbol{c}$, extracted by G_ϕ.



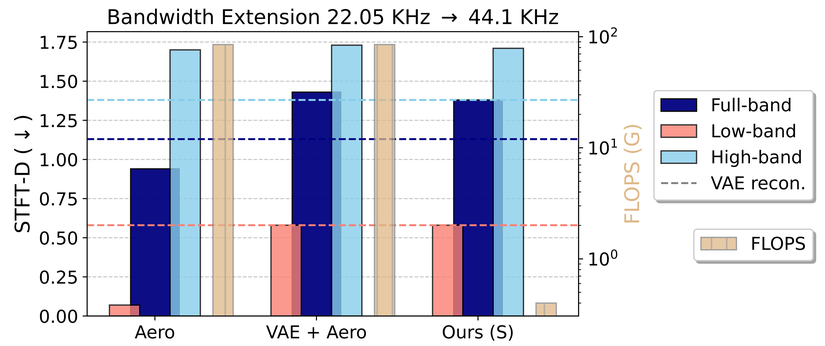
Experiments & Results

Setup

- **Autoencoder**: VAE, 44.1 kHz→ 64 channels at 43 Hz.
- **Latent model** F: Small (S) 4.3 M, Medium (M) 19M params.
  - Training on 1 GPU for <48 hours.
- Raw-audio baseline **evaluation scenarios**:
  1. Undistorted input.
  2. Autoencoded input (VAE + ...).

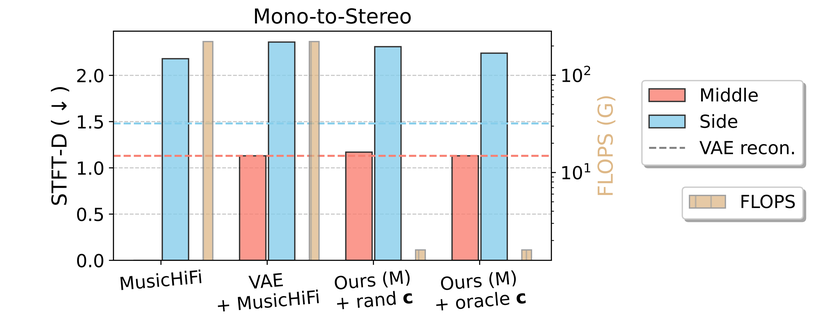
Bandwidth Extension (BWE)

- We showcase the **latent adversarial discriminator** on BWE.
- **Goal**: Upsample a low-band input waveform into a full-band.
- We first perform sinc interpolation and then encode with A_E.



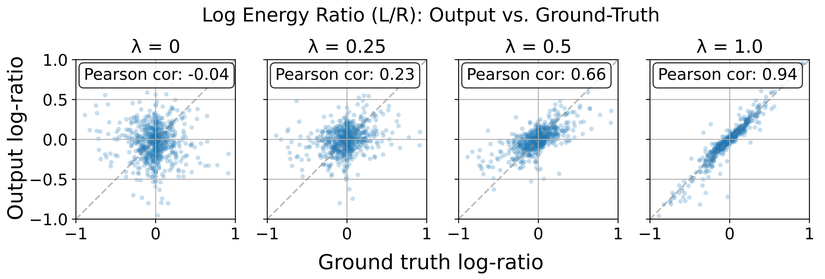
Mono-to-Stereo (M2S)

- We demonstrate the **latent variational encoder** on M2S.
- **Goal**: Upmix mono to stereo with realistic spatialization.



- **Investigating spatial control.** Interpolation on condition $\boldsymbol{c}$.

$$\boldsymbol{c} = \lambda \boldsymbol{c}_{\text{gt}} + (1 - \lambda)\boldsymbol{c}_0,\ \boldsymbol{c}_0 \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$$



Conclusion

- A framework for efficient audio processing in latent spaces.
- **Simplified training** with a latent L_1 loss, augmented with a latent discriminator, or a latent condition encoder.
- Yields significant **efficiency gains**, maintains output quality.
- Natural integration into autoencoder pipelines.

Demos: `re-encoders.github.io/latent-bwe-m2s/`